NIPUN RAUTELA

Software Engineer | Backend & Distributed Systems
nipunrautela.me | github.com/nipunrautela | linkedin.com/in/nipunrautela
rautelanipun@gmail.com | Bengaluru, India | +91-9717104014

EXPERIENCE

Visa Oct 2025 - Present
Software Engineer *Bengaluru*

- **Re-architected** an Ansible-based **automation orchestration** platform **serving 25,000 executions/day** from a polling-driven **Activiti BPMN monolith** into **segregated event-driven processors** on self-healing active-active Hazelcast queues, **cutting execution overhead 60%**.
- **Designed** and **implemented** a **Parking Lot** system for **in-flight migration of all running executions** with **state reconstruction** facilitating **zero-downtime deployment** of event-driven system.
- **Designed & implemented** a **rate limiter** based on **token bucket**, **scoped at User/Global** level, **enforceable via annotation**, and **overridable via API**, which **eliminated DoS by internal applications** and facilitated elevated access for select users.
- **Developed** a **cross-region dual-DC job scheduler** moving out of a custom misfire-prone quartz implementation, **eliminating system-level misfires** and **reducing trigger lag from ±20s to ±5s**.
- Built an AWX simulator with **DSL-driven scenario routing** (DB ops, delays, API responses), **enabling regression and performance testing at 50 executions/min** without loading production Ansible.
- Layered a **tool-calling LLM agent over the simulator**, exposing **custom native tools** that let engineers author simulations from natural language, cutting simulation authoring from **30 minutes to under 5**.

Societe Generale Global Solution Centre Feb 2024 - Oct 2025
Software Engineer *Bengaluru*

- Incorporated API based data-fetching in Java Spring Boot application for compliance, **decomposing intricate SQL queries** to redirect partial data source, while keeping the **response time overhead below 5%**.
- Optimized DB scripts in **C-shell** and **pl/pgSQL** through **parallelization** of independent business logic, **reducing runtime by 45%** (from 5.5 to 3 hours).
- Architected multiple **on-prem deployed** applications on **Azure Public Cloud**, setting up **AKS, Azure PostgreSQL DB, DNS, Certificates, Key Vaults**, etc. and designed automation, **reducing deployment effort by 80%**.
- **Automated blue-green AKS migration** end to end; resource migration, network redirection, and access re-provisioning; **reducing per-cluster cutover** from **4 hours to 1 hour**.
- Led migration of **all** Docker images from **Mirantis Secure Registry to Harbor** and wrote **Jenkins Scripts & Docker files** for updated workflow.

EDUCATION

B.Tech in Computer Science, Vellore Institute of Technology Sep 2020 - Aug 2024
CGPA: 8.99/10

SKILLS

Languages: Java, Python, C++, JavaScript (ES6), SQL, Bash, pl/pgSQL

Backend & Frameworks: Spring Boot, FastAPI, Express.js, REST APIs, Microservices, Activiti BPMN, Quartz Scheduler, JUnit 5, Maven

Distributed Systems & Architecture: System Design, Event-Driven Architecture, Multi-Region & Active-Active Architecture, Rate Limiting, Distributed Job Scheduling, Hazelcast, Ehcache

Databases & Analytics: PostgreSQL, MySQL, ClickHouse

Cloud & DevOps: Microsoft Azure (AKS, Key Vault, PostgreSQL), Docker, Kubernetes, Linux, Ansible (AWX), Jenkins, GitHub Actions, CI/CD, JFrog, Git, Grafana, uv

AI Engineering: LLM Tool/Function Calling, Agentic Workflows, Natural-Language Interfaces, Anthropic API